Can Dr. Whitmore read it aloud?

An agentic briefing tool for FDA CDER drug-shortage testimony

Deloitte Federal FDE take-home #3 · Neil Sahai · November 2026

The incident, replayed on the data you sent

"Is the amoxicillin shortage resolved?"

drug_shortages.csv · line 3 · DS-2023-0142

Resolved revised 2023-11-15

drug_shortages.csv · line 2 · DS-2023-0142

Current revised 2024-01-10

Same record. Two statuses. 56 days apart.

A row scan that finds a Resolved row

"Yes."

Reached a draft of Senate testimony. Correction letter followed.

This system

"No."

Latest revision is Current. The earlier Resolved record is disclosed, not ignored. Tier 1 Certified. Every value cites its row. The conflict warning attached itself — nobody asked for it.

That conflict is in the file you sent. So are fourteen other traps — docs/RED_TEAM_REPORT.md.

Semantic decisions moved from runtime to modeling time

Last year's tool failed because meaning was decided inside a model, invisibly, at runtime. This one defines it once in an ontology you can read, makes numbers unable to pass through the model, labels every claim with a trust tier, and ships with the evals that prove it.

A workflow where it can be, an agent only where it must

Gate

Haiku

Answerable? Ambiguous terms? Before any data.

Planner

Sonnet → Opus on a rule

Declares interpretations first. Resolves literals. Certified tools; SQL last.

Narrator

Sonnet

Placeholders and citations only. Cannot write a number.

Verifier

Opus, always

Atomic claims × evidence × source table. One revision.

Tier

Code

1 Certified · 2 Derived · 3 Qualified · 4 Not answerable

Underneath: DuckDB with a line number on every row · 13-rule data audit whose findings attach to any answer touching a flagged row · ontology.yaml + DECISIONS.md read by the gate, the planner, the narrator and the verifier as citable [SYS] evidence.

Where the model is not used: arithmetic, joins, dedup, fiscal years, FEI checks, coverage, the tier decision. Python, 50 tests, the same code the evals run.

Every answer says how much weight it can bear

1

Certified

Certified tools or a certified query. Verifier passed. Ambiguity resolved or pinned.

Eligible for testimony with sign-off

2

Derived

LLM-authored SQL: guarded, executed, three candidates compared.

Internal use; certify after review

3

Qualified

A reading changes the answer, shown side by side — or a verifier concern.

Pin a reading → re-run → Tier 1

4

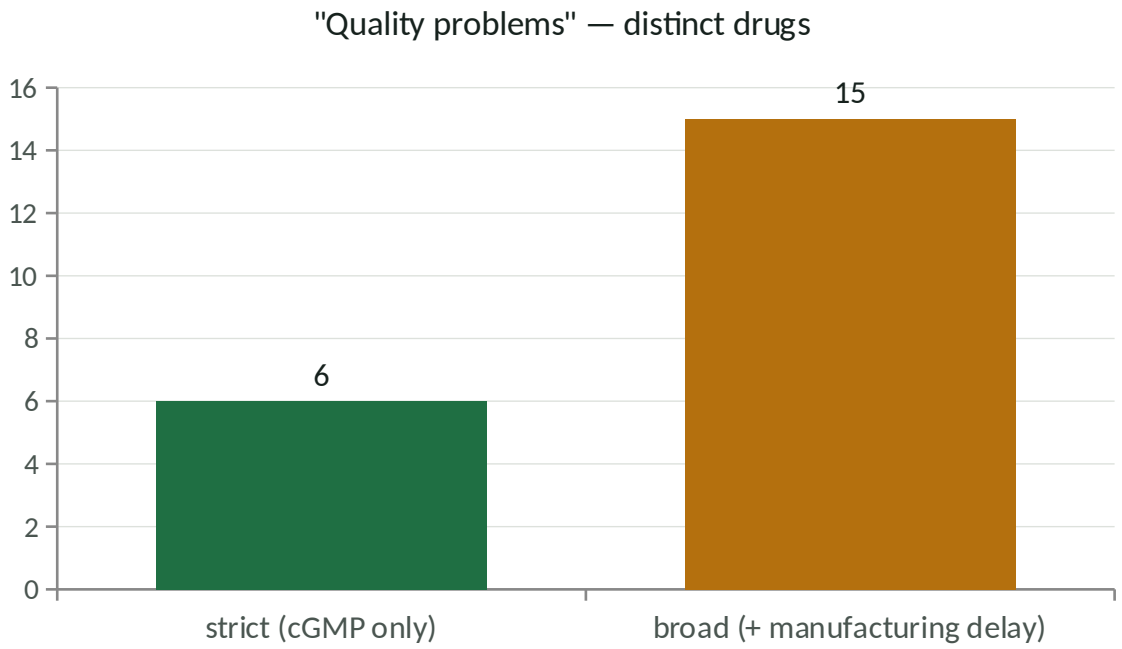
Not answerable

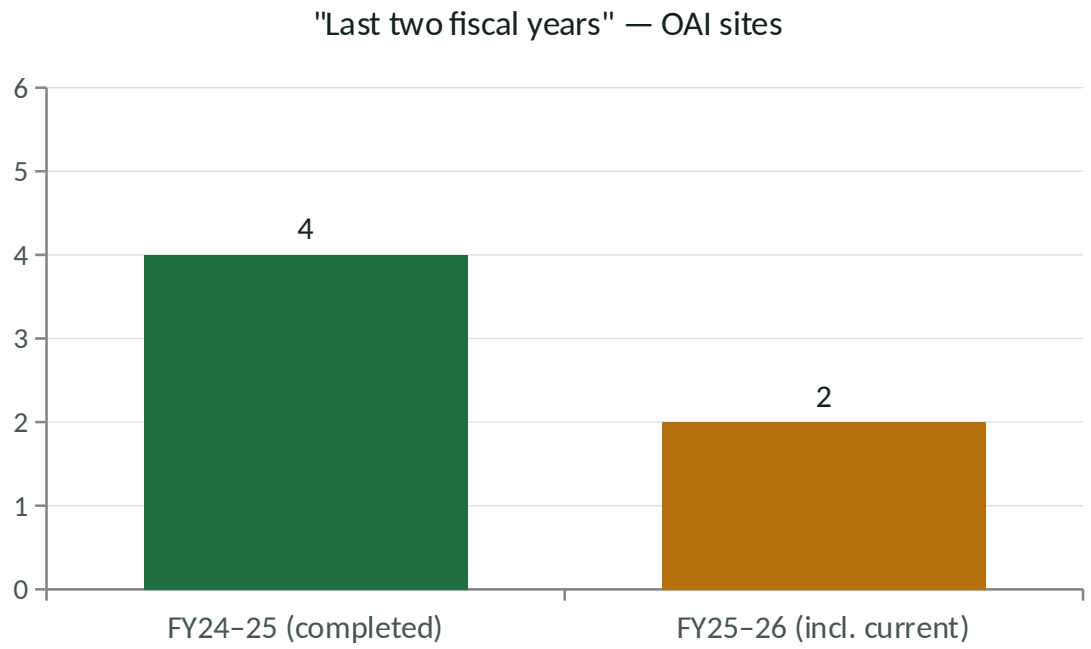
Out of scope, or the data lacks the field.

Says what can be answered instead

A Tier 3 answer with the right numbers is still not testimony until a person decides the reading. The ambiguity is not hidden; it is a one-click decision with a paper trail.

Readings that change the answer are computed, not chosen





Each certified query computes its alternate readings deterministically and reports whether they differ. The trust tier never depends on whether the planner felt like checking. Default readings and their rationale: DECISIONS.md D1–D14; the client confirms or overrides them in week one.

The loop that turns a prototype into a product

1

Off-script question

Tier 2 Derived — guarded SQL, three candidates agree

2

Analyst reviews

Evidence rows, verifier claims, interpretations

3

Certify

Plan + args + pinned readings + who + when + version

4

Next time

Runner executes the verified plan first → Tier 1

5

Backlog

Questions asked but not yet certified = the pharmacist's queue

Same pattern as Snowflake's verified-query repository and Databricks' trusted assets. The three required queries ship this way — verified plans, not hardcoded answers — so they are rigorous without being special-cased, and the panel can see them in the registry next to anything certified during the demo. Every promote and revoke lands in an audit log.

Measured, not claimed

0.0%

answers containing a number not present in evidence (33 runs)

395 / 1

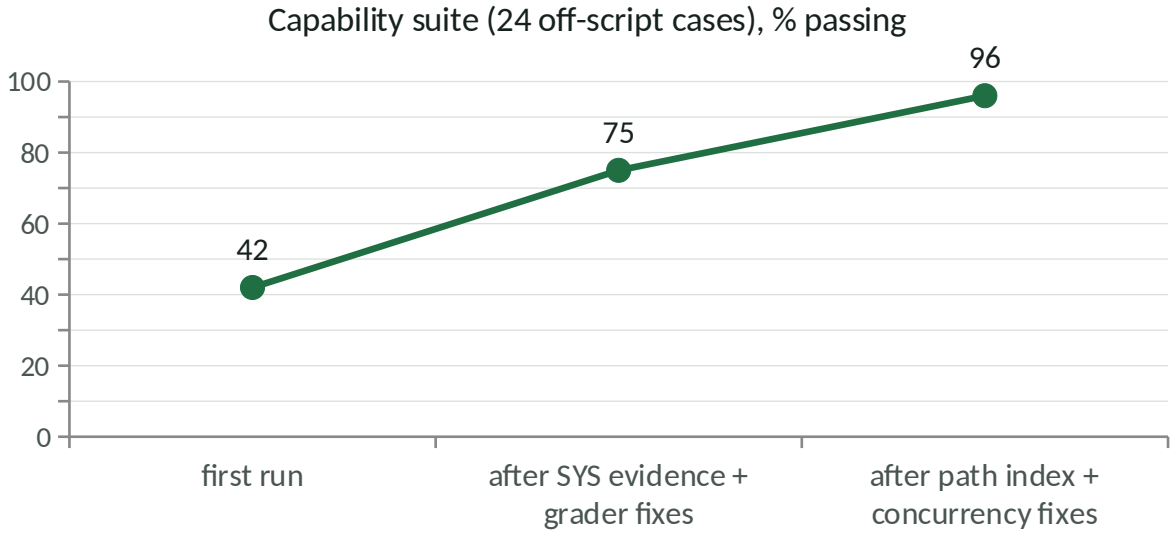
claims verified / flagged after revision, capability suite

10 / 10

adversarial cases passing: fake FEI, fake drug, blame, testimony language, injection…

$0.20–0.29

per question at Anthropic list prices, Opus verification included



Two suites, three grader types

Regression (9 cases): the required queries under default and alternate readings, every planted trap. 8–9/9; the miss is the Opus verifier flagging a framing nit on a correct answer → Tier 3. The safe direction.

Capability (24): off-script, definitional, temporal, adversarial. Code graders for numbers and citations; one model judge per dimension. pass^k, not pass@k.

The verifier caught something on 17–33% of first drafts. That is the cost of the Opus call earning itself.

Where it still fails — said before you ask

Correct number, wrong frame

The renderer cannot catch a true count attached to the wrong noun. The verifier usually does; when it does not, a Certified answer is wrong. That is why sign-off exists and why pass^k is reported.

The join is disclosed, not fixed

44 of 57 current shortages cannot be linked to a site in this snapshot. The system says so in the same sentence as the count. Only the production application registry closes it.

Sixty to ninety seconds

Two narration passes and an Opus verification per question. Right for briefing prep; wrong for a live hearing. Certified queries can be pre-computed before the day.

The ontology is a guess until week one

Defaults follow FDA's public definitions. If ODSS means something narrower by "resolved" or "foreign," every answer is consistently wrong the same way — one file, versioned, sixteen questions to fix it.

From prototype to a hearing: 30 / 60 / 90

Days 1–30 · Shadow mode

- Two analysts run real questions alongside the Excel process; nothing goes in a binder
- Week one: 16 discovery questions rewrite the ontology in ODSS's own definitions
- Staff pharmacist certifies the first ~20 queries from real committee questions

Days 31–60 · Internal briefings

- Tier 1 + signed items into internal briefings for the Deputy Center Director
- Real sources replace the snapshot: shortage DB, Drugs@FDA, FEI registry; nightly audit with diffs
- Claude on Bedrock in AWS GovCloud; nothing leaves the agency account

Days 61–90 · First external memo

- One externally facing memo built entirely from certified, signed items
- Capability suite re-seeded from the pilot backlog; pass^k gates every change
- Decide on scale: which other CDER offices share the sources, what their ontology deltas are

Reusable across offices and agencies: the pipeline, trust tiers, certified registry, eval harness, briefing builder, MCP exposure. Client-specific and isolated: ontology.yaml, DECISIONS.md, the certified tools over the client's tables, the audit rules.

Appendix — the three required queries, as of 2025-10-23

Query	Default reading	Alternate readings computed
1 · Quality shortages / OAI sites	6 drugs (Quality: cGMP Issue). 4 linked sites with an OAI in FY2024–25: Baxter Deerfield, Pfizer McPherson, ICU Medical Lake Forest, Fresenius Kabi Wilson. Hospira: VAI only. Meropenem site: no inspection record, country unknown.	Broad quality scope: 15 drugs / 21 records. FY25–26: 2 sites. Classification-date basis: same 4. "Early 2025" per the PDF: FY23–24, only Pfizer and ICU Medical.
2 · Amoxicillin oral suspension	6 applications for the ingredient (no dosage-form column): 4 fully approved, 2 tentative, 3 actively marketed, 1 discontinued. One oral-suspension record Current (DS-2023-0142, earlier Resolved revision disclosed); one Discontinued.	Include tentative: 6 approved. As of 2023-12-01: DS-2023-0142 was Resolved; conflict still disclosed. Amox-clav records mislinked to an amoxicillin ANDA — flagged.
3 · Foreign sites not inspected since FY2021	4: Aurobindo Unit IV (FY2018), Zhejiang Tianyu (FY2020), Macleods (FY2017), Shanxi Zhendong (FY2019). 2 never-inspected sites reported separately with UNKNOWN country. Coverage: 13 of 57 current records link to a site.	Exclusive reading (< FY2021): same 4. Since FY2024: adds Aurobindo Hyderabad. Two FEIs with a prefix anomaly flagged for verification.